CS3081 Cloud Computing

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Project: Knowledge Distillation & Cloud Deployment of a Lightweight Language Model.

(1) What I Learned?

→ Through this project, I gained practical experience in knowledge distillation, language-model fine-tuning, containerization, kubernetes orchestration, automated CI/CD pipelines and cloud networking. I learned how a larger teacher model can generate structured training data for smaller student model and how the student can be evaluated & deployed as a scaleable API.

(2) Challenges Faced and Solutions

→ One of my biggest challenges was the limited computational resources available on my MacBook Air M2.
Although, the M2 is capable of development workloads, running multiple Docker containers, a kubernetes cluster, object storage, language-models, and load testing tools simultaneously would create substantial memory & storage pressure.

→ Instead of allowing local hardware limitations to compromise the experiments, I adopted a remote execution strategy.

→ I used Google Colab with a T4 GPU for model fine-tuning and Github Actions with Ubuntu runners for automated container builds, kubernetes deployments, CI/CD validation and performance testing.

→ This approach separated development from resource-intensive execution. I could write and maintain the project locally while using disposable remote environments to execute & verify the infrastructure.

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→ Github Actions also improved reproducibility because the deployment steps were defined as version-controlled workflows rather than depending on the configuration of my personal laptop.

However, this was an infrastructure adaption, not exact replacement for testing on a local machine. The performance measurements reflect the Github Actions environment.

③ Key Findings

→ I generated 3,000 teacher-labelled records and retained 2763 after cleaning the dataset.
The student model was then fine tuned using 1500 records.

→ The primary student model achieved 100% JSON validity & 83% intent accuracy on 100 validation examples, although its full-frame accuracy was only 40%.

→ During kubernetes load testing, the service processed 365 requests with zero failures, achieving approximately 3.07 requests/second, and 2.9 seconds P95 latency.

I also verified replica load balancing, rolling updates, rollback, pod recovery & the automated quality gate.

④ Improvements and Future Scope

→ The project could be improved by implementing the remaining object-storage integration, expanding the training dataset, improving full-frame prediction accuracy, and completing additional failure-recovery and autoscaling experiments.

Future testing on dedicated local or cloud kubernetes environment would provide more comprehensive evidence of performance and resource consumption.

* Github Repository:

https://github.com/imnehal360/cs3081-cloud-mini-project

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